Simulation Arena

Approval Bottleneck: Where Deals Actually Get Stuck

CF Approval is the jam: 84.0% busy, adding 2.8 days of waiting, and it is why 29.0% of requests miss the 10-day promise. $1.70M of revenue is stuck behind it. Auto-approve requests under 10% would release $1.05M of that.

Seed 42 | Run run_beb0b9c870ad | 4,000 scenarios tested | Discrete-event queueing simulation (arr… | 22 Sep 2026, 05:05

Same seed, same numbers - this run can be reproduced exactly.

How this works

We rebuilt your approval process as a queue. Requests arrive the way yours do, each waits for a free approver, gets worked on for a realistic amount of time, then moves to the next step - and a request that waits too long sometimes gives up and walks. Running thousands of requests through that queue shows where the pile-up forms, how long the whole journey takes, and how many requests would still be sitting there when the quarter closes. We then re-ran the same queue with each fix in place to price them against each other. Every option is run on the same dice - same arrivals, same requests - so the difference between two options is the fix itself rather than luck.

The numbers that matter

HOW LONG A REQUEST TAKES TODAY
6.7 days
Half of all requests clear in 6.7 days or less.
vs promise is 10 days

CLEARED INSIDE THE 10-DAY PROMISE
71.0%
The share of requests that still land in the quarter they were meant to.
vs 29.0% miss it

WHERE IT JAMS
CF Approval
This step is 84.0% busy and adds 2.8 days of waiting - the single biggest cause of the delay.

BUSIEST STEP IS BUSY
84.0%
Above about 85% a queue stops clearing: any hiccup becomes a backlog.
vs 2.8 days of waiting

REVENUE STUCK BEHIND THE QUEUE
$1.70M
Deal value of the requests that miss the quarter, times the 15.0% you told us is lost when a deal slips.
vs $39.15M of deals pass through in a quarter

WALK-AWAY RATE
0.0%
Requests that waited more than 25 days and gave up.

BEST FIX: AUTO-APPROVE REQUESTS UNDER 10%
-2.5 days
Median time goes from 6.7 to 4.3 days and requests inside the promise go from 71.0% to 88.9%.
vs $1.05M of revenue released

How we got there

1. The process we rebuilt

12.0 requests a week arrive and pass through Deal Desk Review -> CF Approval -> Legal, with 3, 2, 2 approvers respectively. Built from your assumptions, because no history was mapped.

Evidence: $39.15M of deals through the queue per quarter

2. Where it jams

CF Approval is 84.0% busy and makes requests wait 2.8 days on average - 42.5% of the whole journey is spent sitting in front of that one step.

Evidence: 84.0% busy, 2.8 days waiting

3. What that costs

Only 71.0% of requests clear inside the 10-day promise, so 29.0% slip out of the quarter. At 15.0% of deal value lost per slip, that is $1.70M of revenue at risk. Nothing gives up outright, but the slowest tenth wait 15 days.

Evidence: $1.70M at risk

4. What the fixes are worth

Auto-approve requests under 10% is the best of them: median time falls from 6.7 to 4.3 days, requests inside the promise rise from 71.0% to 88.9%, and $1.05M of revenue comes back into the quarter.

Evidence: $1.05M released

5. The bottom line

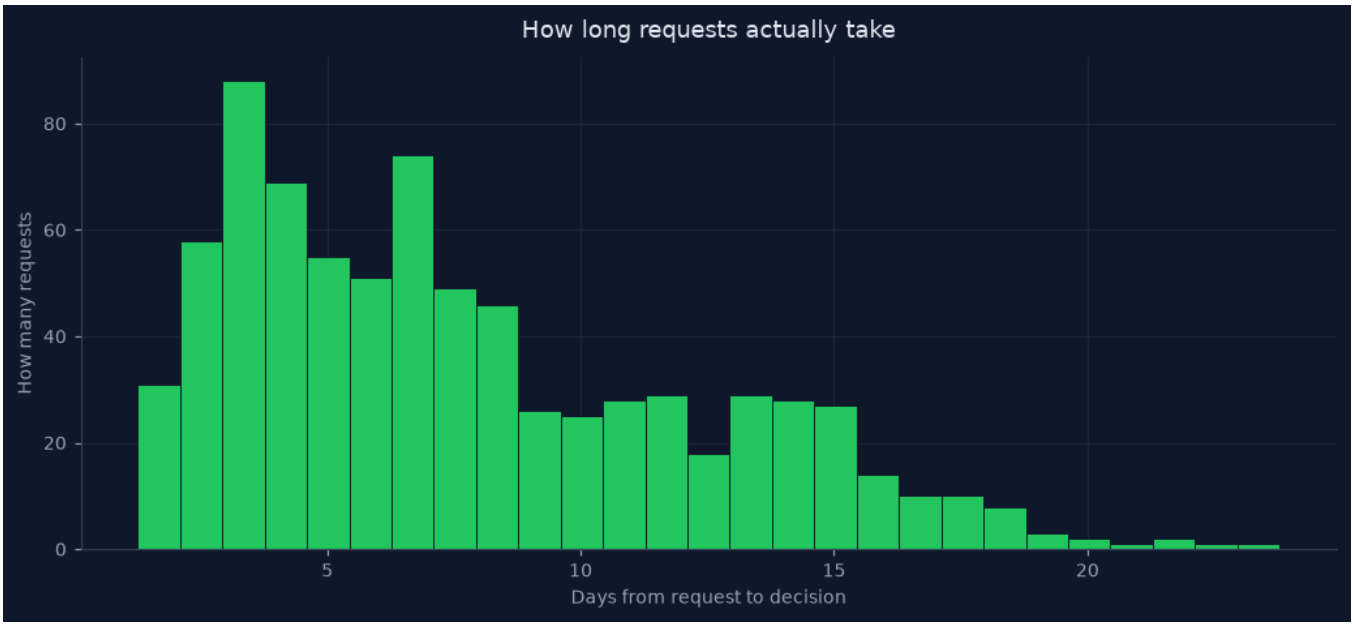
The constraint is not effort, it is capacity at CF Approval: the work itself takes 1.1 days, the waiting takes 2.8. Add capacity or take work out of that queue - asking people to work faster will not move the number.

Evidence: 1.1 days work, 2.8 days waiting

Charts

How long requests actually take

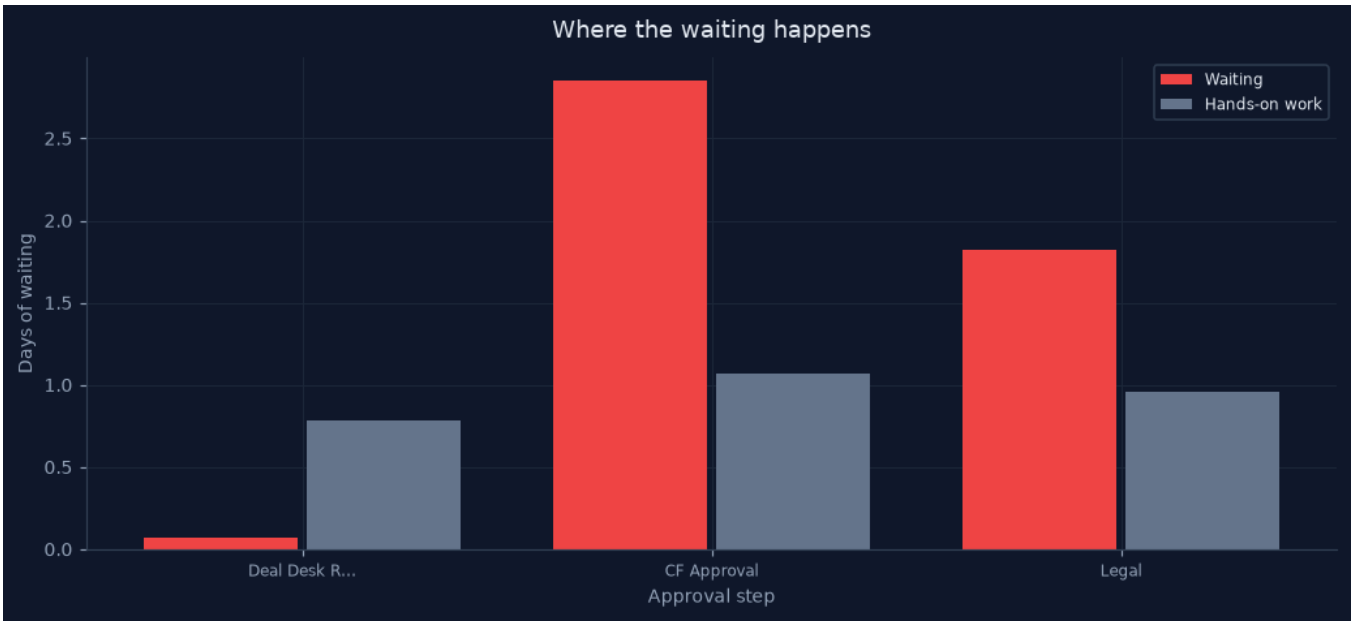
Today's process; the promise is 10 days



What this shows: Everything to the right of the promise line is a deal that slips out of the quarter. The long tail on the right is the queue, not the work.

Where the waiting happens

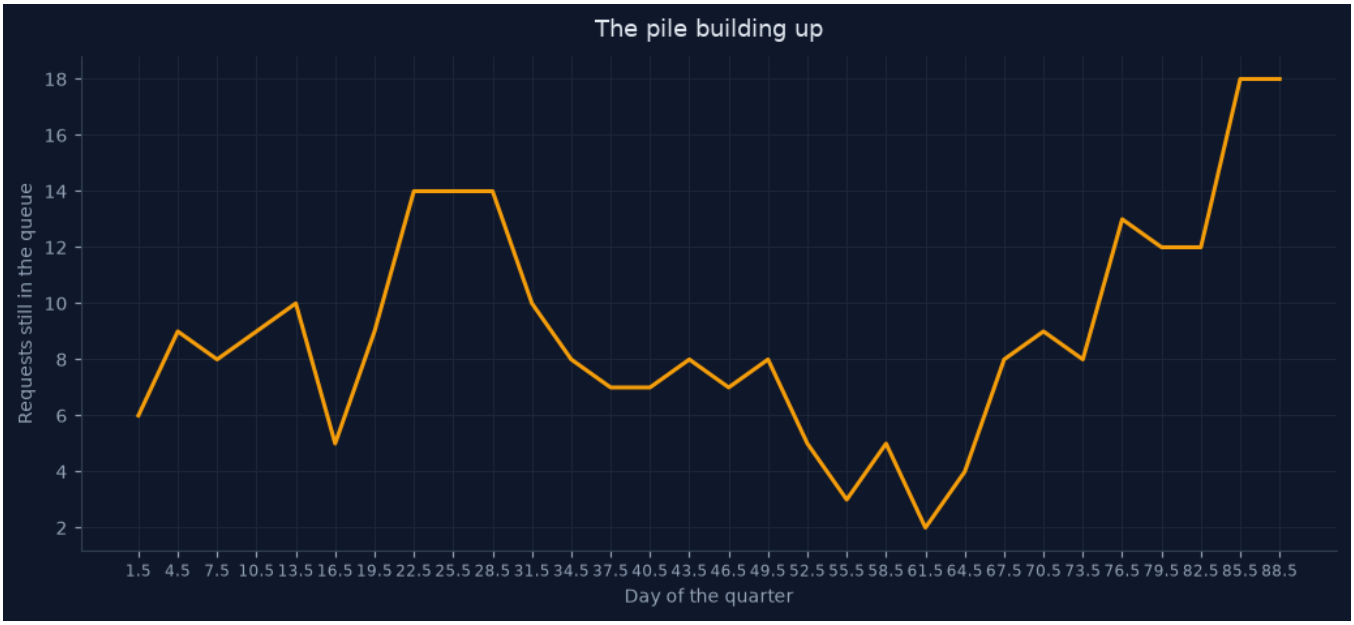
Average days a request sits before each step even starts



What this shows: The tall red bar is the bottleneck. Grey is the actual work - if red dwarfs grey, the problem is capacity, not effort.

The pile building up

Requests open at any point in the quarter



What this shows: A line that keeps climbing means the queue never catches up - that is the definition of a bottleneck.

Tables

Each step, measured

Waiting is the queue; hands-on is the actual work. Share of waiting is where your delay lives.

Step	Approvers	Requests	Avg wait (days)	Slowest 10% wait	Longest wait	Hands-on (days)	How busy	Share of all waiting	Column 10
Deal Desk Review	3	783	0.1	0.2	1.4	0.8	44.9%	1.4%	-
CF Approval	2	783	2.8	5.6	16.2	1.1	84.0%	60.1%	BOTTLENECK
Legal	2	783	1.8	5.5	10.5	1.0	73.2%	38.4%	-

The bottleneck step is the one with the most waiting per unit of busyness - it is the reason the quarter slips.

What each fix is worth

Revenue at risk = deal value of the requests that miss the 10-day promise, times 15.0%.

Option	Days to clear (median)	Slowest 10% (days)	Inside the promise	Walk-away rate	Busiest step busy	Revenue at risk	Gain
Today's process	6.7	14.6	71.0%	0.0%	84.0%	$1.70M	-
One more approver in CF Approval	5.2	12.4	85.3%	0.0%	76.2%	$862k	$840k released
Auto-approve requests under 10%	4.3	10.5	88.9%	0.0%	77.5%	$652k	$1.05M released
One more approver in Legal	5.0	11.0	84.9%	0.0%	84.0%	$885k	$818k released

The highlighted row is the cheapest fix that releases the most revenue.

The ten worst-hit requests

The requests that lost the most time in today's process.

Request	Arrived (day)	Days to clear	Of which waiting	Stuck behind	Deal value	Value at risk	Outcome
Request 30	21	23.8	0.7	-	$250k	$38k	Late
Request 165	87	22.8	10.0	CF Approval	$250k	$38k	Late
Request 128	76	21.8	6.6	CF Approval	$250k	$38k	Late
Request 166	89	21.4	13.9	CF Approval	$250k	$38k	Late
Request 64	33	21.0	7.2	CF Approval	$250k	$38k	Late
Request 172	90	19.9	17.5	CF Approval	$250k	$38k	Late
Request 171	90	19.8	17.8	CF Approval	$250k	$38k	Late
Request 170	90	19.5	16.7	CF Approval	$250k	$38k	Late
Request 163	86	19.3	16.4	CF Approval	$250k	$38k	Late
Request 161	86	18.9	15.8	CF Approval	$250k	$38k	Late

Multiply this pattern across the quarter and you have the size of the problem.

What we recommend

Fix CF Approval first. Auto-approve requests under 10% releases $1.05M of revenue and takes 2.5 days out of the process.

Risk Low 29 out of 100 | Confidence: Medium

Why

• CF Approval runs at 84.0% busy and adds 2.8 days of waiting - more than any other step.

• Only 71.0% of requests clear inside the 10-day promise, putting $1.70M of revenue at risk.

• The work in that step takes 1.1 days; the waiting takes 2.8. It is a capacity problem, not an effort problem.

What to do

1. Auto-approve requests under 10% - worth $1.05M of revenue released and 2.5 days off the median.

2. Auto-approve anything under 10% discount without a CF Approval stop - that alone removes 40.0% of the queue.

3. Publish a daily view of the CF Approval queue with a 10-day clock on each request, so slippage is visible before the quarter closes.

4. Watch the tail: the slowest requests take 15 days. Nothing is walking away yet, but at that length the customer has already re-planned without you.

The trade-off: Adding an approver costs salary; auto-approving below a threshold costs a little margin on small deals. Both are far cheaper than the $1.70M currently at risk.

What we assumed and what to watch

Worth knowing before you act

• No approval history was mapped, so the queue is built from your assumptions. Upload the request log (request id, submitted day, stage, days in stage, approver) to measure the real thing.

Assumptions

• No usable history: 12 requests a week, 90 working days in the quarter.

• Hands-on time per step: Deal Desk Review about 0.8 days; CF Approval about 1.1 days; Legal about 0.9 days

• Times are skewed: most requests are quicker than average, a few take much longer.

• Approvers per step: Deal Desk Review 3, CF Approval 2, Legal 2 (from your settings)

• A request that waits more than 25 days gives up with 25.0% probability.

• Revenue at risk = 15.0% of the deal value of every request that misses the 10-day promise; average deal value $250,000.

• Every request is modelled as passing through every step; steps that only some requests reach therefore look slightly busier than they are.

• 5 simulated quarters of 200 requests each per option (1,000 requests in total); about 156 arrive and clear inside each 90-day quarter. Seed 42, so re-running gives identical numbers. Every option is tested against the same simulated requests, so the comparison is fair. Revenue at risk is the average per quarter.

• Request / ticket ID -> your 'deal_id' column

• Day it was submitted -> not in your file; not used

• Approval step -> your 'stage' column

• Days it spent in that step -> your 'days_elapsed_total' column

• Who approved it -> not in your file; not used

• Deal value -> your 'amount' column

Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

Time	Stage	What happened
0.00s	prepare	Found 75,069 rows and 16 columns.
0.00s	prepare	You chose 'deal_id' for 'Request / ticket ID'.
0.00s	prepare	We could not find a 'Day it was submitted' column. We left it out and will work without it.
0.00s	prepare	You chose 'stage' for 'Approval step'.
0.00s	prepare	You chose 'days_elapsed_total' for 'Days it spent in that step'.
0.00s	prepare	We could not find a 'Who approved it' column. We left it out and will work without it.
0.00s	prepare	You chose 'amount' for 'Deal value'.
0.00s	prepare	Request / ticket ID -> your 'deal_id' column
0.00s	prepare	Day it was submitted -> not in your file; not used
0.00s	prepare	Approval step -> your 'stage' column
0.00s	prepare	Days it spent in that step -> your 'days_elapsed_total' column
0.00s	prepare	Who approved it -> not in your file; not used
0.00s	prepare	Deal value -> your 'amount' column
1.53s	prepare	Read 75,069 records into the model.
1.53s	simulate	Running the simulation…

Time	Stage	What happened
1.53s	read	Reading how approval requests arrive and how long each step takes...
1.94s	prepare	No usable approval history, so we built the process from your assumptions: Deal Desk Review -> CF Approval -> Legal.
1.94s	prepare	Rebuilding the queue: 12.0 requests a week, 3 in Deal Desk Review, 2 in CF Approval, 2 in Legal.
1.94s	simulate	Running 1,000 requests through the queue (5 simulated quarters)...
6.25s	summarize	Found it: CF Approval is the jam - 84.0% busy and 2.8 days of waiting.
6.47s	simulate	Tested 'One more approver in CF Approval'.
6.61s	simulate	Tested 'Auto-approve requests under 10%'.
6.73s	simulate	Tested 'One more approver in Legal'.
6.73s	summarize	Comparing the fixes...
6.73s	summarize	Done. CF Approval is the bottleneck - $1.70M of revenue is stuck behind it.